Week 9: Sales and Customer Analytics System

Scenario

The e-commerce company analyzes sales data to understand customer purchasing behavior, product performance, and revenue trends. Reports help management identify top customers, best-selling products, and profitable categories.

Average Order Value

Definition: Average order value is the mean amount spent per order. AVG calculates the mean of order totals, and ROUND formats the result to two decimal places.

```
SQL> SELECT COUNT(*) AS Number_Of_Orders_Placed FROM Orders;

NUMBER_OF_ORDERS_PLACED
-----------------------
                     14
```

Highest and Lowest Order Amount

Definition: MAX returns the largest order amount, while MIN returns the smallest order amount in the Orders table.

```
SQL> SELECT SUM(TotalAmount) AS Total_Revenue FROM Orders;

TOTAL_REVENUE
-------------
       206200
```

Orders per Customer

Definition: COUNT counts each customer's orders. LEFT JOIN keeps customers who have not placed orders, so they can also appear in the report.

```
SQL> SELECT ROUND(AVG(TotalAmount),2) AS Average_Order_Value FROM Orders;

AVERAGE_ORDER_VALUE
-------------------
           14728.57
```

Orders per Customer

Definition: COUNT counts each customer's orders. LEFT JOIN keeps customers who have not placed orders, so they can also appear in the report.

```
SQL> SELECT MAX(TotalAmount) AS Highest_Order,
  2  MIN(TotalAmount) AS Lowest_Order
  3  FROM Orders;

HIGHEST_ORDER LOWEST_ORDER
------------- ------------
        55850         2450
```

Orders per Customer

Definition: COUNT counts each customer's orders. LEFT JOIN keeps customers who have not placed orders, so they can also appear in the report.

```
SQL> SELECT SUM(TotalAmount) AS Total_Revenue FROM Orders
  2  WHERE OrderDate BETWEEN DATE '2026-06-01' AND DATE '2026-08-31';

TOTAL_REVENUE
-------------
        37450
```

Orders per Customer

Definition: COUNT counts each customer's orders. LEFT JOIN keeps customers who have not placed orders, so they can also appear in the report.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS TOTAL_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_I CUSTOMER_NAME                          TOTAL_ORDERS
---------- ---------------------------------------- ------------
CUST000001 Arjun Kumar                                       1
CUST000015 Nithin Murugan                                    1
CUST000003 Rahul Menon                                       1
CUST000004 Ananya Iyer                                       1
CUST000005 Vikram Reddy                                      1
CUST000006 Meera Krishnan                                    1
CUST000007 Suresh Nair                                       1
CUST000008 Kavya Rajan                                       1
CUST000009 Karthik Subramanian                               1
CUST000010 Divya Pillai                                      1
CUST000011 Aditya Verma                                      1

CUSTOMER_I CUSTOMER_NAME                          TOTAL_ORDERS
---------- ---------------------------------------- ------------
CUST000012 Sneha Patel                                       1
CUST000002 Priya Sharma                                      1
CUST000014 Lakshmi Venkatesan                                1
CUST000013 Rohan Bose                                        0

15 rows selected.
```

Total Amount Spent by Each Customer

Definition: SUM calculates each customer's total spending. NVL replaces a NULL total with zero for a customer who has no purchases.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      NVL(SUM(O.TOTALAMOUNT), 0) AS TOTAL_AMOUNT_SPENT
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUSTOMER_I CUSTOMER_NAME                          TOTAL_AMOUNT_SPENT
---------- ---------------------------------------- ------------------
CUST000001 Arjun Kumar                                        55850
CUST000014 Lakshmi Venkatesan                                 28000
CUST000003 Rahul Menon                                        28000
CUST000005 Vikram Reddy                                       22000
CUST000009 Karthik Subramanian                                18500
CUST000015 Nithin Murugan                                      8350
CUST000004 Ananya Iyer                                         8300
CUST000011 Aditya Verma                                        7000
CUST000006 Meera Krishnan                                      7000
CUST000007 Suresh Nair                                         6500
CUST000008 Kavya Rajan                                         5700

CUSTOMER_I CUSTOMER_NAME                          TOTAL_AMOUNT_SPENT
---------- ---------------------------------------- ------------------
CUST000010 Divya Pillai                                        5200
CUST000002 Priya Sharma                                        3350
CUST000012 Sneha Patel                                         2450
CUST000013 Rohan Bose                                             0

15 rows selected.
```

Customers with Maximum Purchase Amount

Definition: This query compares customer spending totals and returns the customer or customers whose total equals the highest total.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      ROUND(NVL(AVG(O.TOTALAMOUNT), 0), 2) AS AVERAGE_SPENDING
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY AVERAGE_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                          AVERAGE_SPENDING
---------- ---------------------------------------- ----------------
CUST000001 Arjun Kumar                                        55850
CUST000014 Lakshmi Venkatesan                                 28000
CUST000003 Rahul Menon                                        28000
CUST000005 Vikram Reddy                                       22000
CUST000009 Karthik Subramanian                                18500
CUST000015 Nithin Murugan                                      8350
CUST000004 Ananya Iyer                                         8300
CUST000011 Aditya Verma                                        7000
CUST000006 Meera Krishnan                                      7000
CUST000007 Suresh Nair                                         6500
CUST000008 Kavya Rajan                                         5700

CUSTOMER_I CUSTOMER_NAME                          AVERAGE_SPENDING
---------- ---------------------------------------- ----------------
CUST000010 Divya Pillai                                        5200
CUST000002 Priya Sharma                                        3350
CUST000012 Sneha Patel                                         2450
CUST000013 Rohan Bose                                             0

15 rows selected.
```

Customers with Fewer Purchases

Definition: COUNT calculates the number of orders per customer. Sorting the counts helps identify customers with fewer purchases.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_PURCHASE_AMOUNT
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING SUM(O.TOTALAMOUNT) = (
 13      SELECT MAX(TOTAL_SPENDING)
 14      FROM (
 15          SELECT SUM(TOTALAMOUNT) AS TOTAL_SPENDING
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  );

CUSTOMER_I CUSTOMER_NAME                              TOTAL_PURCHASE_AMOUNT
---------- ---------------------------------------- ---------------------
CUST000001 Arjun Kumar                                               55850
```

Top 5 Customers by Total Spending

Definition: Customer orders are grouped by customer, SUM calculates total spending, and descending order places the highest spenders first. FETCH FIRST 5 ROWS ONLY limits the result to five customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) <= 1
 13  ORDER BY NUMBER_OF_ORDERS;

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ---------------
CUST000013 Rohan Bose                                             0
CUST000015 Nithin Murugan                                         1
CUST000003 Rahul Menon                                            1
CUST000004 Ananya Iyer                                            1
CUST000005 Vikram Reddy                                           1
CUST000006 Meera Krishnan                                         1
CUST000007 Suresh Nair                                            1
CUST000008 Kavya Rajan                                            1
CUST000009 Karthik Subramanian                                    1
CUST000010 Divya Pillai                                           1
CUST000011 Aditya Verma                                           1

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ---------------
CUST000012 Sneha Patel                                            1
CUST000014 Lakshmi Venkatesan                                     1
CUST000001 Arjun Kumar                                            1
CUST000002 Priya Sharma                                           1

15 rows selected.
```

Top 5 Customers by Total Spending

Definition: Customer orders are grouped by customer, SUM calculates total spending, and descending order places the highest spenders first. FETCH FIRST 5 ROWS ONLY limits the result to five customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_SPENDING DESC
 13  FETCH FIRST 5 ROWS ONLY;

CUSTOMER_I CUSTOMER_NAME                                TOTAL_SPENDING
---------- ---------------------------------------- ---------------
CUST000001 Arjun Kumar                                           55850
CUST000003 Rahul Menon                                           28000
CUST000014 Lakshmi Venkatesan                                    28000
CUST000005 Vikram Reddy                                          22000
CUST000009 Karthik Subramanian                                   18500
```

Customers with Maximum Number of Orders

Definition: The query calculates each customer's order count and compares it with the maximum order count to identify the highest-frequency customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) = (
 13      SELECT MAX(ORDER_COUNT)
 14      FROM (
 15          SELECT COUNT(*) AS ORDER_COUNT
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  )
 20  ORDER BY CUSTOMER_NAME;

CUSTOMER_I CUSTOMER_NAME                            NUMBER_OF_ORDERS
---------- ---------------------------------------- ---------------
CUST000011 Aditya Verma                                            1
CUST000004 Ananya Iyer                                             1
CUST000001 Arjun Kumar                                             1
CUST000010 Divya Pillai                                            1
CUST000009 Karthik Subramanian                                     1
CUST000008 Kavya Rajan                                             1
CUST000014 Lakshmi Venkatesan                                      1
CUST000006 Meera Krishnan                                          1
CUST000015 Nithin Murugan                                          1
CUST000002 Priya Sharma                                            1
CUST000003 Rahul Menon                                             1

CUSTOMER_I CUSTOMER_NAME                            NUMBER_OF_ORDERS
---------- ---------------------------------------- ---------------
CUST000012 Sneha Patel                                             1
CUST000007 Suresh Nair                                             1
CUST000005 Vikram Reddy                                            1

14 rows selected.
```

High-Value Customers

Definition: High-value customers are those whose total spending exceeds a chosen threshold. SUM calculates spending and HAVING filters the grouped results.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) >= 2
 13  ORDER BY NUMBER_OF_ORDERS DESC;

no rows selected
```

High-Value Customers

Definition: High-value customers are those whose total spending exceeds a chosen threshold. SUM calculates spending and HAVING filters the grouped results.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS,
  5      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMERID
  9  GROUP BY
 10      C.CUSTOMER_ID,
 11      C.FIRST_NAME,
 12      C.LAST_NAME
 13  HAVING SUM(O.TOTALAMOUNT) > 20000
 14  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
TOTAL_SPENDING
--------------
CUST000001 Arjun Kumar                                           1
        55850

CUST000014 Lakshmi Venkatesan                                    1
        28000

CUST000003 Rahul Menon                                           1
        28000


CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
TOTAL_SPENDING
--------------
CUST000005 Vikram Reddy                                          1
        22000
```

Products with Maximum Sales Quantity

Definition: SUM of ordered quantities calculates units sold for each product. Sorting the totals from highest to lowest identifies best-selling products.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT PRODUCT_NAME                                       TOTAL_QUANTITY_SOLD
------- -------------------------------------------------- -------------------
P000006 USB Cable                                                            6
P000005 Smartphone                                                           6
P000015 Wireless Earbuds                                                     3
P000008 Smart Watch                                                          2
P000012 External Hard Disk                                                   2
P000009 Headphones                                                           2
P000001 Laptop                                                               1
P000013 Gaming Monitor                                                       1
P000010 Tablet                                                               1
P000014 Laptop Stand                                                         1
P000002 Wireless Mouse                                                       1

PRODUCT PRODUCT_NAME                                       TOTAL_QUANTITY_SOLD
------- -------------------------------------------------- -------------------
P000003 Mechanical Keyboard                                                  1
P000004 Bluetooth Speaker                                                    1

13 rows selected.
```

Products Generating Highest Revenue

Definition: Revenue is calculated by multiplying the ordered quantity by the product price and summing those values for each product.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  5      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  6  FROM PRODUCT P
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      P.PRODUCT_ID,
 11      P.PRODUCT_NAME
 12  ORDER BY TOTAL_REVENUE DESC;

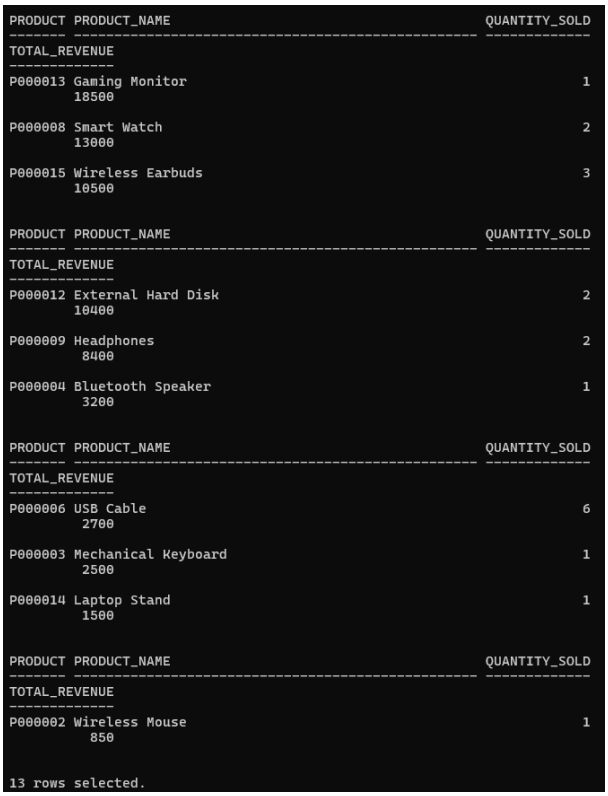
PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000005 Smartphone                                                     6
       168000

P000001 Laptop                                                         1
        55000

P000010 Tablet                                                         1
        22000
```

Best-Selling Product Revenue Results

Definition: This continuation displays sales quantities and revenue for additional products, allowing product performance to be compared.



```
PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000013 Gaming Monitor                                                 1
        18500

P000008 Smart Watch                                                    2
        13000

P000015 Wireless Earbuds                                               3
        10500


PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000012 External Hard Disk                                             2
        10400

P000009 Headphones                                                     2
         8400

P000004 Bluetooth Speaker                                              1
         3200


PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000006 USB Cable                                                      6
         2700

P000003 Mechanical Keyboard                                            1
         2500

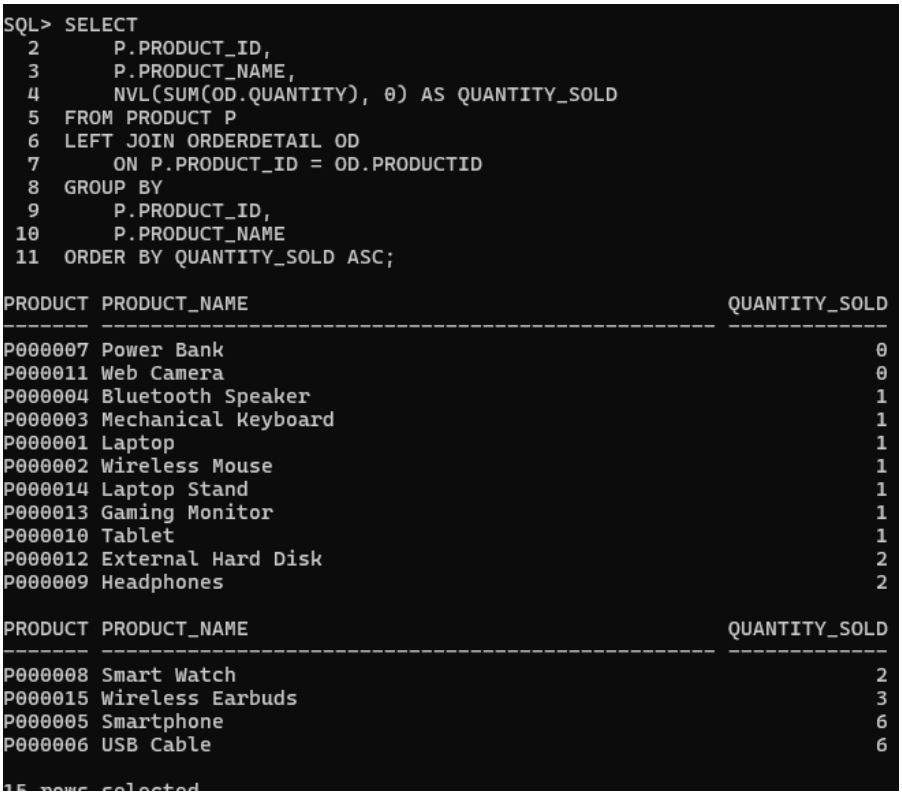
P000014 Laptop Stand                                                   1
         1500


PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000002 Wireless Mouse                                                 1
          850


13 rows selected.
```

Least-Selling Products

Definition: Products are sorted by total quantity sold in ascending order. LEFT JOIN includes products with no sales, and NVL displays their missing quantity total as zero.



```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  5  FROM PRODUCT P
  6  LEFT JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY QUANTITY_SOLD ASC;

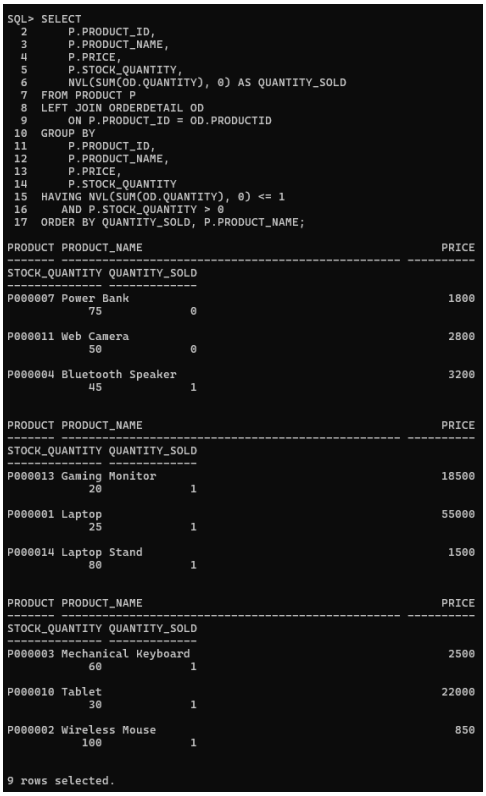
PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
P000007 Power Bank                                                     0
P000011 Web Camera                                                     0
P000004 Bluetooth Speaker                                              1
P000003 Mechanical Keyboard                                            1
P000001 Laptop                                                         1
P000002 Wireless Mouse                                                 1
P000014 Laptop Stand                                                   1
P000013 Gaming Monitor                                                 1
P000010 Tablet                                                         1
P000012 External Hard Disk                                             2
P000009 Headphones                                                     2

PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
P000008 Smart Watch                                                    2
P000015 Wireless Earbuds                                               3
P000005 Smartphone                                                     6
P000006 USB Cable                                                      6

15 rows selected.
```

Products Requiring Promotion

Definition: Products with little or no sales relative to their stock may require promotion to increase customer interest and sales.



```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      P.PRICE,
  5      P.STOCK_QUANTITY,
  6      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  7  FROM PRODUCT P
  8  LEFT JOIN ORDERDETAIL OD
  9      ON P.PRODUCT_ID = OD.PRODUCTID
 10  GROUP BY
 11      P.PRODUCT_ID,
 12      P.PRODUCT_NAME,
 13      P.PRICE,
 14      P.STOCK_QUANTITY
 15  HAVING NVL(SUM(OD.QUANTITY), 0) <= 1
 16     AND P.STOCK_QUANTITY > 0
 17  ORDER BY QUANTITY_SOLD, P.PRODUCT_NAME;

PRODUCT PRODUCT_NAME                                       PRICE
------- -------------------------------------------------- ----------
STOCK_QUANTITY QUANTITY_SOLD
-------------- -------------
P000007 Power Bank                                          1800
            75             0

P000011 Web Camera                                          2800
            50             0

P000004 Bluetooth Speaker                                   3200
            45             1


PRODUCT PRODUCT_NAME                                       PRICE
------- -------------------------------------------------- ----------
STOCK_QUANTITY QUANTITY_SOLD
-------------- -------------
P000013 Gaming Monitor                                     18500
            20             1

P000001 Laptop                                             55000
            25             1

P000014 Laptop Stand                                        1500
            80             1


PRODUCT PRODUCT_NAME                                       PRICE
------- -------------------------------------------------- ----------
STOCK_QUANTITY QUANTITY_SOLD
-------------- -------------
P000003 Mechanical Keyboard                                 2500
            60             1

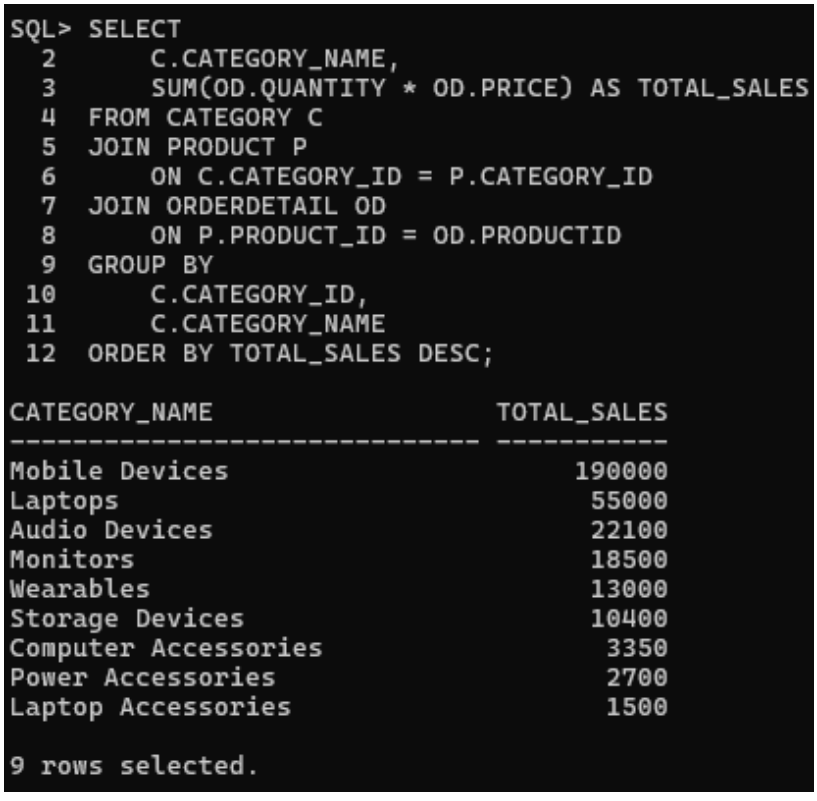
P000010 Tablet                                             22000
            30             1

P000002 Wireless Mouse                                       850
           100             1


9 rows selected.
```

Total Sales by Category

Definition: The query joins categories, products, and order details, then sums quantity multiplied by price to calculate total revenue for each category.



```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY TOTAL_SALES DESC;

CATEGORY_NAME                  TOTAL_SALES
------------------------------ -----------
Mobile Devices                      190000
Laptops                              55000
Audio Devices                        22100
Monitors                             18500
Wearables                            13000
Storage Devices                      10400
Computer Accessories                  3350
Power Accessories                     2700
Laptop Accessories                    1500

9 rows selected.
```

Number of Products Sold in Each Category

Definition: SUM of order quantities calculates the number of units sold in each category, allowing categories to be compared by sales volume.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY) AS PRODUCTS_SOLD
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY PRODUCTS_SOLD DESC;

CATEGORY_NAME                  PRODUCTS_SOLD
------------------------------ -------------
Mobile Devices                             7
Power Accessories                          6
Audio Devices                              6
Computer Accessories                       2
Wearables                                  2
Storage Devices                            2
Laptop Accessories                         1
Laptops                                    1
Monitors                                   1

9 rows selected.
```

Highest Revenue-Generating Category

Definition: The query calculates revenue for each category and identifies the category with the greatest total revenue.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
 13      SELECT MAX(CATEGORY_REVENUE)
 14      FROM (
 15          SELECT SUM(OD2.QUANTITY * OD2.PRICE) AS CATEGORY_REVENUE
 16          FROM CATEGORY C2
 17          JOIN PRODUCT P2
 18              ON C2.CATEGORY_ID = P2.CATEGORY_ID
 19          JOIN ORDERDETAIL OD2
 20              ON P2.PRODUCT_ID = OD2.PRODUCTID
 21          GROUP BY C2.CATEGORY_ID
 22      )
 23  );

CATEGORY_NAME                  TOTAL_REVENUE
------------------------------ -------------
Mobile Devices                        190000
```

Average Sales per Category

Definition: This query calculates the average sales value for each category to compare category-level performance.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      ROUND(AVG(OD.QUANTITY * OD.PRICE), 2) AS AVERAGE_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY AVERAGE_SALES DESC;

CATEGORY_NAME                  AVERAGE_SALES
------------------------------ -------------
Mobile Devices                         95000
Laptops                                55000
Monitors                               18500
Wearables                               6500
Storage Devices                         5200
Audio Devices                        3683.33
Computer Accessories                    1675
Laptop Accessories                      1500
Power Accessories                        900

9 rows selected.
```

Sales Performance Report

Definition: This report summarizes total sales, number of orders, average order value, and the highest order value to provide an overview of sales performance.

```
SQL> SELECT
  2      SUM(TOTALAMOUNT) AS TOTAL_SALES,
  3      COUNT(ORDERID) AS NUMBER_OF_ORDERS,
  4      ROUND(AVG(TOTALAMOUNT), 2) AS AVERAGE_ORDER_VALUE,
  5      MAX(TOTALAMOUNT) AS HIGHEST_ORDER_VALUE
  6  FROM ORDERS;

TOTAL_SALES NUMBER_OF_ORDERS AVERAGE_ORDER_VALUE HIGHEST_ORDER_VALUE
----------- ---------------- ------------------- -------------------
     206200               14            14728.57                55850
```

Customer Analytics Report

Definition: The customer report shows customer names, order counts, total spending, and purchase frequency to describe customer purchasing behavior.

```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS,
  4      NVL(SUM(O.TOTALAMOUNT), 0) AS TOTAL_SPENDING,
  5      COUNT(O.ORDERID) AS PURCHASE_FREQUENCY
  6  FROM CUSTOMER C
  7  LEFT JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMERID
  9  GROUP BY
 10      C.CUSTOMER_ID,
 11      C.FIRST_NAME,
 12      C.LAST_NAME
 13  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Arjun Kumar                                           1          55850
                 1

Lakshmi Venkatesan                                    1          28000
                 1

Rahul Menon                                           1          28000
                 1


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Vikram Reddy                                          1          22000
                 1

Karthik Subramanian                                   1          18500
                 1

Nithin Murugan                                        1           8350
                 1
```

Customer Analytics: Continued Results

Definition: These results continue the customer analytics report by showing spending, order counts, and purchase frequency for additional customers.

```
CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Ananya Iyer                                           1           8300
                 1

Aditya Verma                                          1           7000
                 1

Meera Krishnan                                        1           7000
                 1


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Suresh Nair                                           1           6500
                 1

Kavya Rajan                                           1           5700
                 1

Divya Pillai                                          1           5200
                 1


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Priya Sharma                                          1           3350
                 1

Sneha Patel                                           1           2450
                 1

Rohan Bose                                            0              0
                 0


15 rows selected.
```

Product Performance Report

Definition: The product report groups order details by product and calculates quantity sold and revenue generated.

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS REVENUE_GENERATED
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY REVENUE_GENERATED DESC;
```

Product Performance: Revenue Results

Definition: This continuation lists product sales quantity and revenue, helping identify products contributing most to total sales.

```
PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Smartphone                                                     6
          168000

Laptop                                                         1
           55000

Tablet                                                         1
           22000


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Gaming Monitor                                                 1
           18500

Smart Watch                                                    2
           13000

Wireless Earbuds                                               3
           10500


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
External Hard Disk                                             2
           10400

Headphones                                                     2
            8400

Bluetooth Speaker                                              1
            3200


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
USB Cable                                                      6
            2700

Mechanical Keyboard                                            1
            2500

Laptop Stand                                                   1
            1500
```

Product Performance: Revenue Results

Definition: This continuation lists product sales quantity and revenue, helping identify products contributing most to total sales.

```
PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Wireless Mouse                                                 1
            850


13 rows selected.
```

Category Analysis Report

Definition: This report groups sales by category and displays category name, total quantity sold, and total revenue for comparison.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY) AS TOTAL_PRODUCTS_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5  FROM CATEGORY C
  6  JOIN PRODUCT P
  7      ON C.CATEGORY_ID = P.CATEGORY_ID
  8  JOIN ORDERDETAIL OD
  9      ON P.PRODUCT_ID = OD.PRODUCTID
 10  GROUP BY
 11      C.CATEGORY_ID,
 12      C.CATEGORY_NAME
 13  ORDER BY TOTAL_REVENUE DESC;

CATEGORY_NAME                  TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------------------ ------------------- -------------
Mobile Devices                                   7        190000
Laptops                                          1         55000
Audio Devices                                    6         22100
Monitors                                         1         18500
Wearables                                        2         13000
Storage Devices                                  2         10400
Computer Accessories                             2          3350
Power Accessories                                6          2700
Laptop Accessories                               1          1500

9 rows selected.
```